

BUSINESS REQUIREMENTS DOCUMENT - LKFORUM

Version 1.0

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Status: Draft

1. Objective and Scope

1.1. Project Objective

The primary objective of the LKForum project is to design and build a modern, scalable, and highly interactive community platform. Unlike general social networks, LKForum focuses on **autonomous community management**, allowing users to create, configure, and moderate their own niche sub-forums (Communities).

The development focuses on delivering a high-quality system that ensures:

- High performance for real-time interactions, including Instant Messaging and Notifications.
- Superior scalability to handle a large and growing number of concurrent communities and users.
- Robust moderation capabilities, integrating both automated (AI-assisted) and manual review processes.
- High standards for security and data privacy to protect user information and platform integrity.

1.2. Project Scope

The project scope includes the following modules:

- **Identity & Access Management:** Multi-factor authentication, SSO, and granular role-based access control.
 - **Community Management:** Creation, configuration (public/private/18+), rule-setting, and membership handling.
 - **Content Management:** Multi-format posts (text, image, video, poll), nested comments, and draft management.
 - **Engagement System:** Real-time voting (upvote/downvote), saved posts, and activity history.
 - **Communication Suite:** Real-time 1-1 messaging via WebSockets and an instant notification system.
 - **Moderation & Governance:** Report handling, user banning/muting, and content approval workflows.
 - **Administration Dashboard:** Global system oversight, analytics, and platform-wide moderation.
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2. Business Requirement

2.1. Application Overview

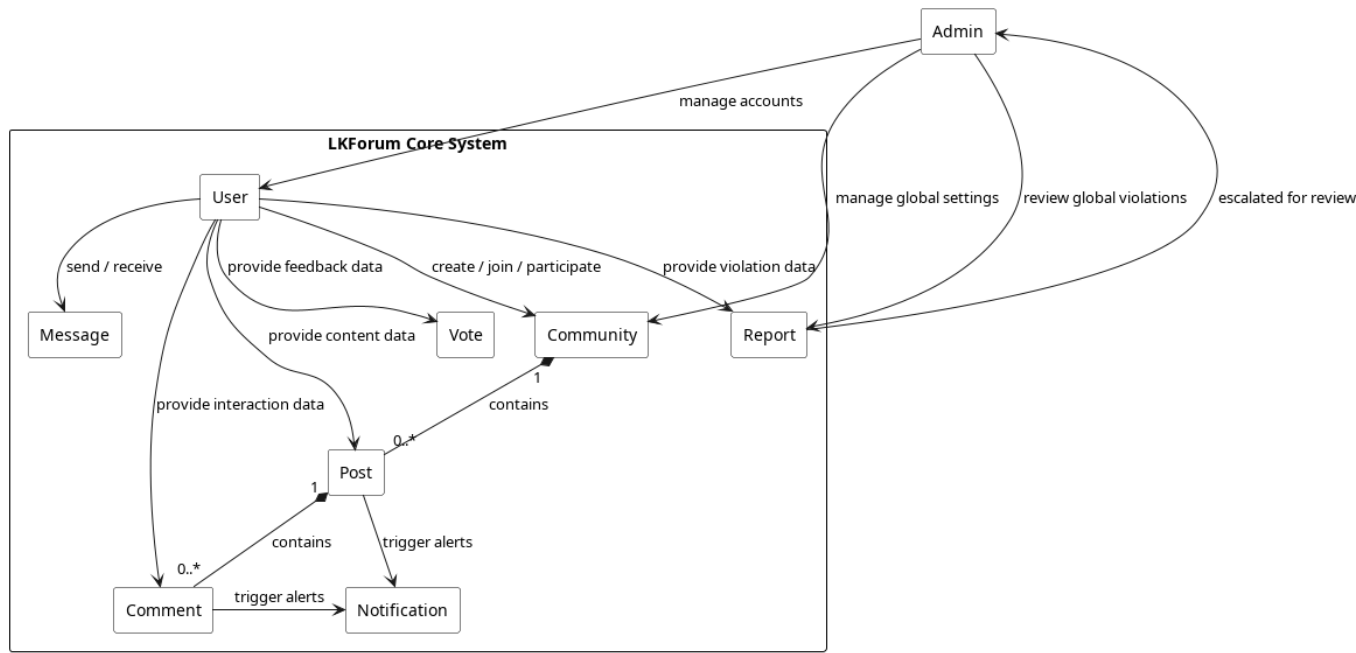
LKForum is a community-driven social network. It is structured around "Communities" (sub-forums).

- **Users** can discover and join communities based on interests.
- **Content** is categorized by communities, ensuring relevant discussions.

- **Decentralized Moderation:** Each community has its own set of rules and a dedicated moderation team (Owners/Mods), reducing the burden on global administrators.
- **Real-time Experience:** The platform feels "alive" through instant feedback, live messaging, and notifications.

2.2. Domain Model

2.2.1. Domain Diagram (Conceptual)



2.2.2. Domain Objects Description

#	Object Name	Object Description
1	User	A registered identity with a profile, karma points, and activity history.
2	Community	A niche hub with specific topics, rules, and privacy settings.
3	Post	Content shared within a community (Text, Image, Video, or Poll).
4	Comment	A response to a post, supporting nested/threaded replies.
5	Vote	An expression of approval (Upvote) or disapproval (Downvote) on content.
6	Message	Direct communication content between users in a 1-1 chat.
7	Notification	Real-time alerts for replies, mentions, votes, or system messages.
8	Report	A user-generated complaint about content or behavior violating rules.

2.3. Workflow (Key Business Processes)

2.3.1. Content Lifecycle & Moderation Workflow

1. **Creation:** A user creates a post in a community.
2. **AI Screening:** The system automatically checks content for prohibited keywords/media.

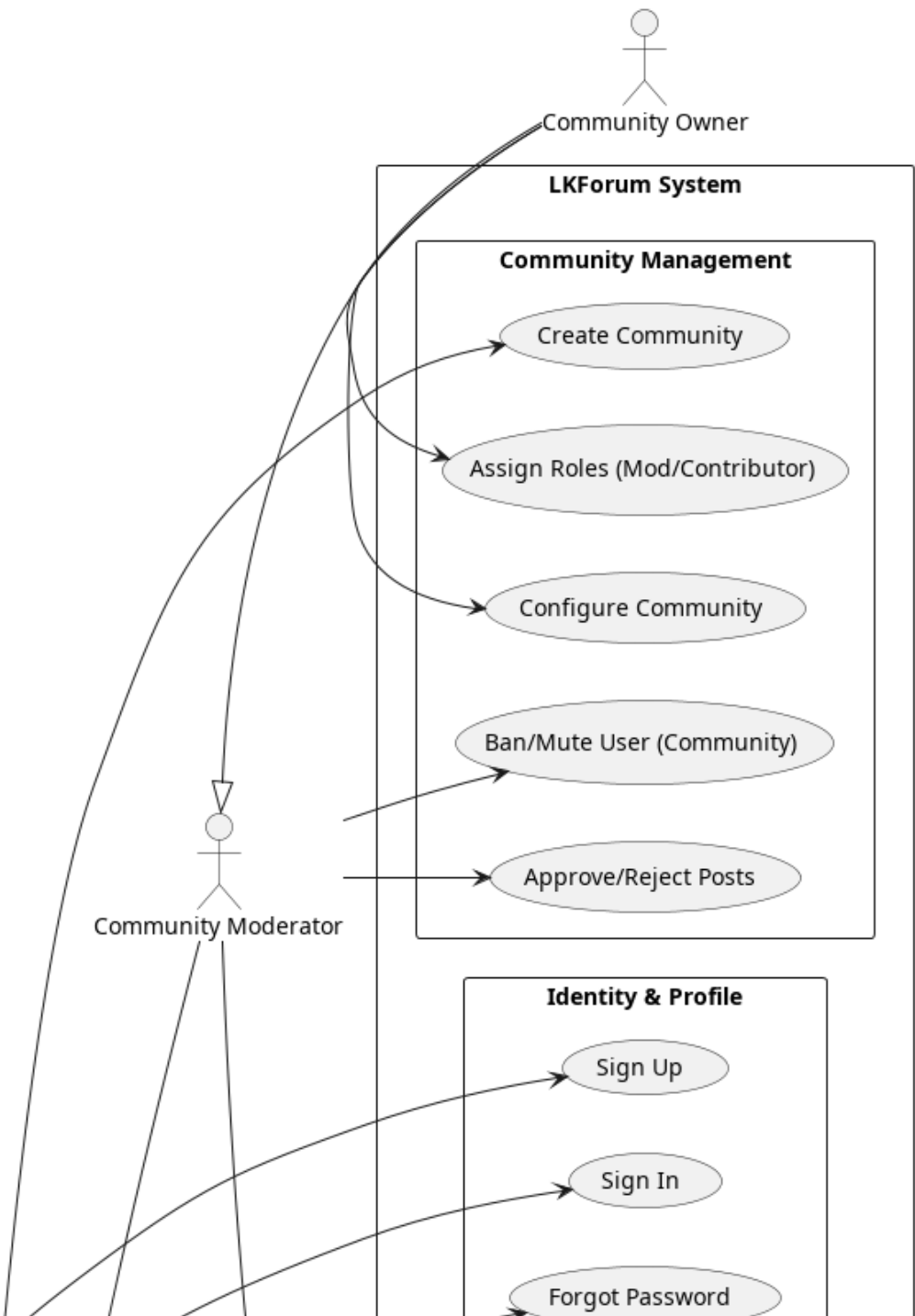
- 3. **Approval (Optional):** If the community requires "Post Approval," the post is held in a *Pending* state.
- 4. **Moderation:** A Moderator/Owner reviews the post (Approve or Reject).
- 5. **Publication:** Once approved (or if no approval is required), the post becomes public and triggers notifications to members.

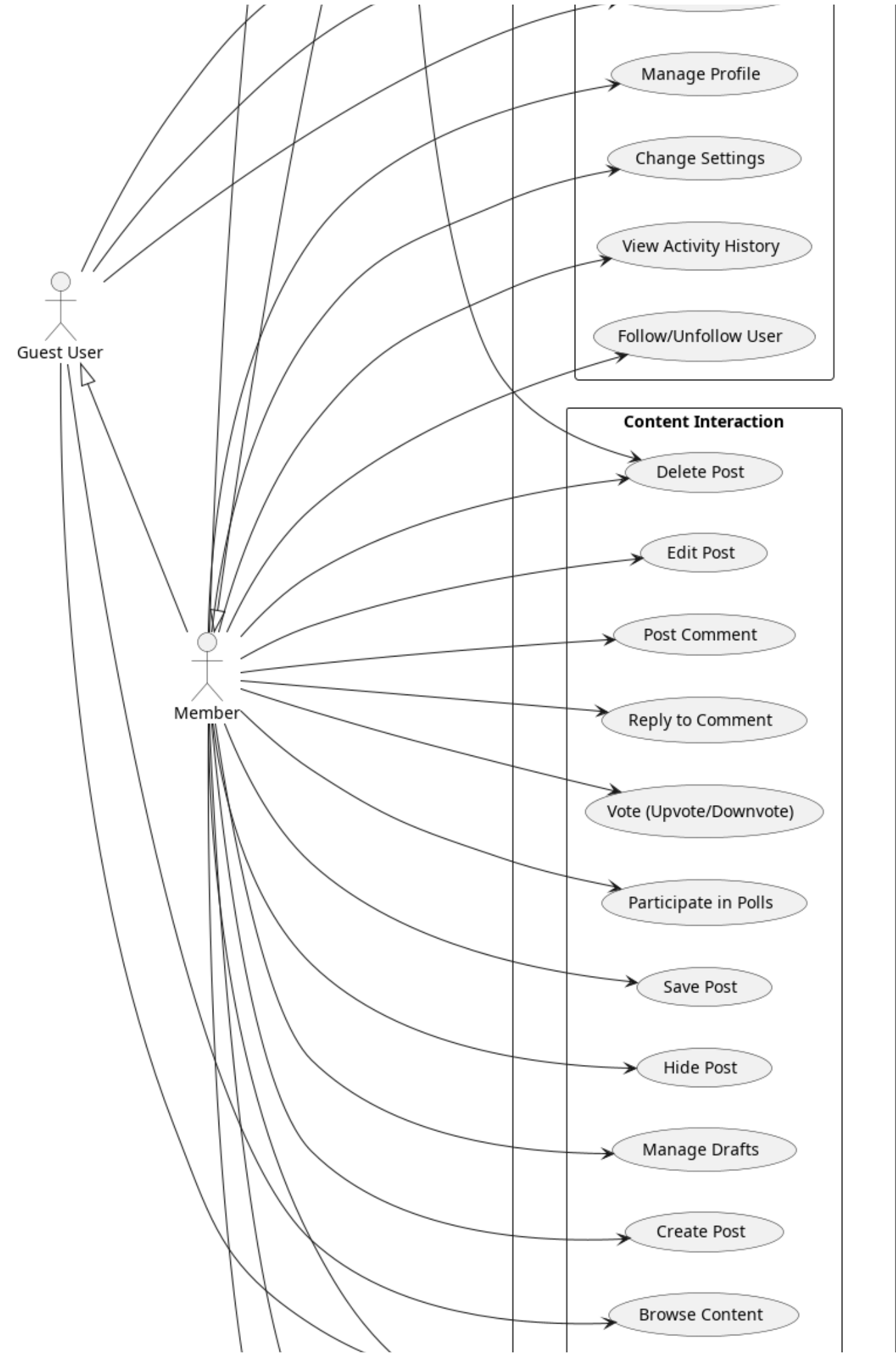
2.3.2. Community Management Workflow

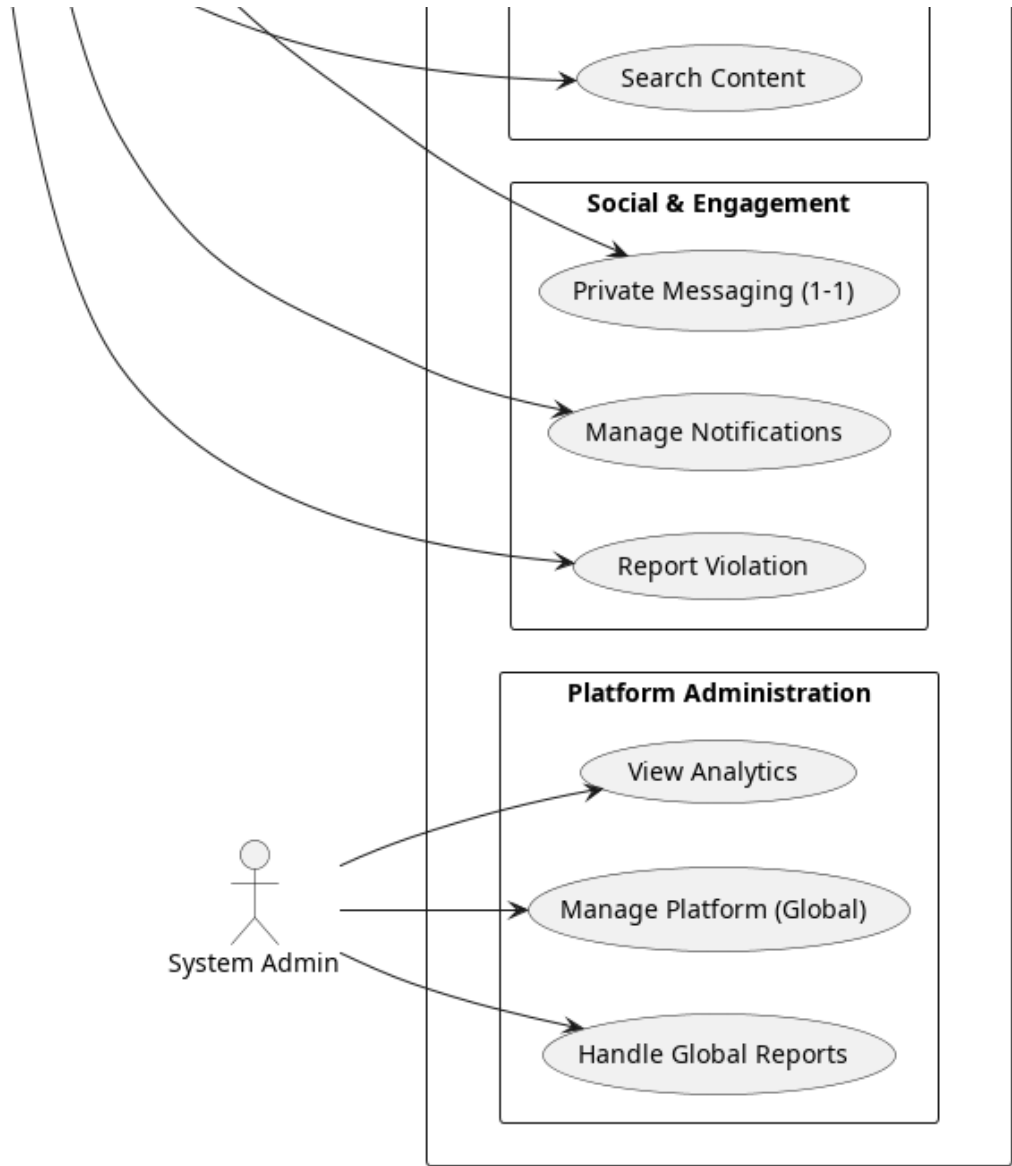
- 1. **Setup:** A user creates a community (becomes the Owner).
- 2. **Configuration:** Owner sets rules, privacy (Public/Private), and invites Moderators.
- 3. **Governance:** Owners/Mods monitor reports and can ban/mute users or remove content.

2.4. Use Cases and Actors

2.4.1. Diagram







2.4.2. Description of Actors

#	Actor Name	Definition
1	Guest User	Unauthenticated user; can browse public content and search.
2	Member	Authenticated user; can post, comment, vote, and message.
3	Community Moderator	Member with management rights in a specific community (Approve/Delete content).
4	Community Owner	Creator of a community; can appoint moderators and change settings.
5	System Admin	Global administrator responsible for platform-wide health and safety.

2.4.3. Description of Use Cases

#	Use Case Name	Definition
1	Sign Up	Users create a new account by providing email, username, and password, verified via OTP.

#	Use Case Name	Definition
2	Sign In	Authenticated access using credentials or linked Google OAuth account.
3	Forgot Password	Users reset their password through email verification (OTP).
4	Manage Profile	Users update personal information, including bio, avatar, and cover image.
5	Change Settings	Users configure account preferences (privacy, notifications, theme).
6	View Activity History	Users review their past posts, comments, and interactions.
7	Follow/Unfollow User	Users subscribe to another user's activity feed.
8	Browse Content	Users explore communities and view public posts without necessarily logging in.
9	Search Content	Users search for specific keywords, posts, users, or communities.
10	Create Post	Members publish new content (Text, Image, Video) within a selected community.
11	Edit Post	Members modify the content of their existing posts (stores edit history).
12	Delete Post	Members or Moderators remove posts from public view (soft delete).
13	Post Comment	Users add a response directly to a post.
14	Reply to Comment	Users respond to an existing comment (supporting nested threads).
15	Vote on Content	Users express approval (Upvote) or disapproval (Downvote) on posts/comments.
16	Participate in Poll	Users cast votes in community-created polls.
17	Save Post	Users bookmark posts for quick access later.
18	Hide Post	Users hide specific posts from their personal feed.
19	Manage Drafts	Users save unfinished posts as drafts and resume editing later.
20	Private Messaging	Users send and receive instant 1-1 messages via WebSocket.
21	Manage Notifications	Users view and clear real-time alerts for engagement and system updates.
22	Report Violation	Users flag inappropriate content or behavior for moderator review.
23	Create Community	Users establish a new niche forum and become the "Owner."
24	Approve/Reject Post	Moderators review pending posts before they appear in the community.
25	Ban/Mute User	Moderators restrict a user's ability to participate within a community.

#	Use Case Name	Definition
26	Configure Community	Owners adjust rules, privacy settings, and appearance of their community.
27	Assign Roles	Owners appoint other members as Moderators or Contributors.
28	Handle Global Reports	System Admins review and act on platform-wide violation reports.
29	Manage Platform	System Admins ban/unban users or communities globally.
30	View Analytics	System Admins monitor platform statistics (DAU, growth, content volume).

2.5. Security Matrix

#	Function	Guest	Member	Mod	Owner	Admin
1	Sign Up / Sign In / Forgot Password	X	X	X	X	X
2	Manage Profile & Settings		X	X	X	X
3	Follow / Unfollow User		X	X	X	X
4	Browse Public Content & Search	X	X	X	X	X
5	View Own Activity History		X	X	X	X
6	Create Post / Comment / Reply		X	X	X	X
7	Edit / Delete Own Content		X	X	X	X
8	Vote / Participate in Polls		X	X	X	X
9	Save / Hide Post / Manage Drafts		X	X	X	X
10	Send/Receive Private Messages		X	X	X	X
11	Receive Notifications		X	X	X	X
12	Report Violations		X	X	X	X
13	Create New Community		X	X	X	X
14	Approve / Reject / Pin Posts			X	X	X
15	Delete Any Post (In Community)			X	X	X
16	Ban / Mute User (In Community)			X	X	X
17	Configure Community Settings				X	X
18	Assign Roles (Mod/Contributor)				X	X
19	Handle Global Reports					X
20	Global User / Community Ban					X
21	View System Analytics					X

2.6. User Story

2.6.1. For Members

- **As a Guest User**, I want to **sign up with my email and verify via OTP** so that I can create a secure account on the platform.
- **As a Member**, I want to **sign in using my Google account** so that I can access the forum quickly.
- **As a Member**, I want to **customize my profile with a bio, avatar, and banner** so that I can express my identity.
- **As a Member**, I want to **save my posts as drafts** so that I can continue editing them later before publishing.
- **As a Member**, I want to **create posts with images and polls** so that I can share rich content.
- **As a Guest User**, I want to **search for specific keywords or communities** so that I can find content that interests me easily.
- **As a Member**, I want to **upvote or downvote posts and comments** so that I can help surface the most valuable content.
- **As a Member**, I want to **reply to specific comments in a threaded view** so that I can maintain structured conversations.
- **As a Member**, I want to **receive real-time notifications for replies and mentions** so that I can stay engaged.
- **As a Member**, I want to **follow other interesting users** so that I can see their latest activities in my feed.
- **As a Member**, I want to **send private messages to other users** so that I can have one-on-one conversations securely.
- **As a Member**, I want to **report inappropriate content** so that I can help maintain a safe community environment.

2.6.2. For Community Owners & Moderators

- **As a Community Owner**, I want to **set specific rules and privacy settings** so that I can define how my community operates.
- **As a Community Moderator**, I want to **review and approve pending posts** so that I can ensure only high-quality content is published.
- **As a Community Owner**, I want to **assign moderator roles to trusted members** so that they can help me manage the community.
- **As a Community Moderator**, I want to **ban or mute disruptive users within my community** so that I can protect other members.

2.6.3. For System Administrators

- **As a System Admin**, I want to **access a global dashboard with system analytics** so that I can monitor the overall health of the platform.
- **As a System Admin**, I want to **handle reports escalated from communities** so that I can take final action on serious violations.
- **As a System Admin**, I want to **perform global bans on malicious users or communities** so that I can keep the entire platform safe.

3. Appendix

3.1. Glossary

- **Karma:** A reputation score based on the net value of upvotes and downvotes received.
- **Sub-forum:** Synonymous with "Community" in LKForum.
- **SSO:** Single Sign-On (Google Authentication).
- **OTP:** One-Time Password sent via email for verification.

3.2. Open Issues

- Implementation of cross-posting (sharing posts across multiple communities).
- Automated AI moderation for image/video content (currently focusing on text).
- Integration of a global search engine (Elasticsearch).